

MARCO SCARLATA

 [linkedin.com/in/marco-scarlata](https://www.linkedin.com/in/marco-scarlata) |  github.com/mscarla2

SKILLS

Languages : C++, Python, Kotlin, Java, JavaScript

Frameworks: gRPC, Protocol Buffers, Boost (Asio), Flask, Django, React, Svelte

Libraries/Tools: Pigweed, CMake, GTest, Tensorflow, TFLite, Keras, Pandas, NumPy, SQL, Git, Bazel/Blaze, LLVM/Clang

EXPERIENCE

Google | Software Engineer

Mountain View, CA

WatchSW AI/ML & Algorithms

Sep 2022 – Present

- Launched **One-Handed Gestures** (Double Pinch & Wrist Turn), a novel, low-power ML gesture recognition algorithm (C++, Python, TFLite) for the Pixel Watch, partnering with a Research Scientist to lead end-to-end integration, on-device optimization, and sensor fusion (IMU & physiological signals); achieved over **92% accuracy**, **<1ms inference**, and **<250ms E2E latency**.
- Architected the **Gesture Hardware Abstraction Layer (HAL)** to ingest and transform sensor events for the platform service, decoupling the algorithm to enable vendor-agnostic implementations; introduced bidirectional communication for context-aware sensing (e.g., display state), facilitating cross-device orchestration with xDevices such as **AR/VR glasses** and reducing latency by **30%** through the consolidation of abstraction layers.
- Owned and optimized the **Low Latency Off-body Detection (LLOB)** algorithm, a core component for most algorithms and system performance, resolving false detections and power regressions, raising reliability across all shipped devices.
- Implemented on-device telemetry and ETL dashboards; led Gestures and LLOB bug triage, closing **2000+ bugs**.
- Mentored teammates on sensor integration and MCU-level development; led design reviews and authored the *"MCU-based Practical Telemetry Guide"* referenced by **24+ engineers**.
- Granted **C++ Readability**, a selective authority that allows one to act as a language reviewer and enforce architectural standards across core production codebases.

Cloud Asset Inventory & Search

- Productionized a scalable C++ solution to enrich Cloud Asset Inventory with structured metadata for cloud-managed assets, impacting **90%+** of GCP resources and boosting adoption to **500,000+** active users.
- Enabled launch of Asset Enrichment in the Asset Query System (C++, SQL, Spanner), powering efficient queries across **275+** GCP asset types for SCC customers.
- Drove cross-team optimization of a workflow runner (gRPC, Borg, Python), reducing development time by **25%** resulting in **10 SWE weeks** saved.

OpenSesame | Software Engineering Intern

Portland, OR

Front-End Division

Jun 2020 – Aug 2020

- Enhanced course language selection with searchable dropdowns (**Angular, TypeScript**); built and ported 10+ end-to-end tests from **Drupal** to **Angular/Selenium**, expanding QA coverage
- Resolved 6+ sprint bugs—including a critical IE landing page issue—by extrapolating burn-down processes and strengthening test automation with Behat API, and unit tests

ANDSystems | Machine Learning Intern

Ulaanbaatar, Mongolia

Machine Learning Team

Jun 2019 – Aug 2019

- Analyzed the purchase history of **100,000+** users buying coupons by regression analysis using **Python, Pandas, NumPy, and PyTorch**, identifying an under-marketed sector in sales that increased revenue by 10%
- Launched an MVP module-based recommender system with caching for an e-commerce platform (*BananaMall*) with 10,000+ downloads on the Playstore and 100,000+ users, using **Python, SKLearn, and DynamoDB**

University of Rochester | Teaching Assistant & CETL Tutor

Rochester, NY

Computer Science Department

Sep 2019 – Dec 2021

- Taught students in *Web Programming, Data Structures & Algorithms, and Formal System & Computations* courses
- Conducted tutoring sessions with 15+ college students, resulting in a 30% increase in their respective course grades

EDUCATION

University of Rochester

Rochester, NY

Bachelors in Computer Science | Minor in Psychology

Aug 2018 – May 2022

- Major GPA:** 3.5/4.0 – Consecutive Dean's List Recipient
- Relevant Courses:** Data Structures & Algorithms | Formal Systems & Computation | Web Programming | Human Computer Interaction | Intro to Artificial Intelligence | Database Systems | Natural Language Processing |
- Dean's Scholarship & Rochester National Grant Awardee